

Brain Drain

Identifying U.S. Regions at Risk for Future Capital Flight Using Social Capital and Mobility Outcomes

Mercy Eme

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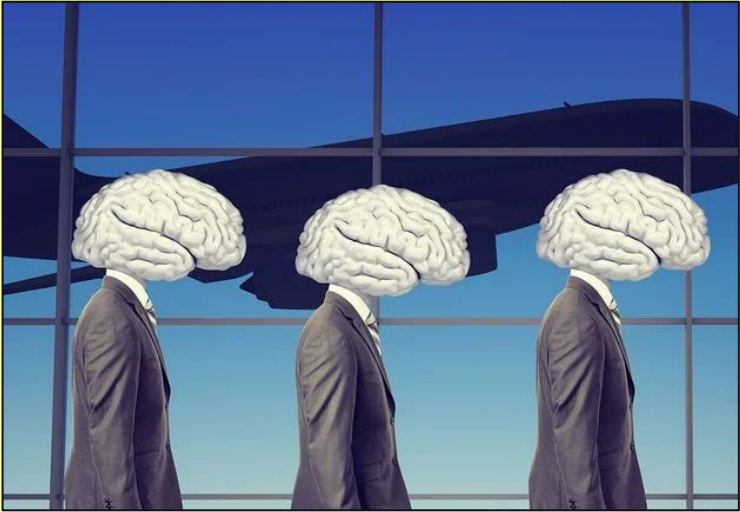
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What is “brain drain?”

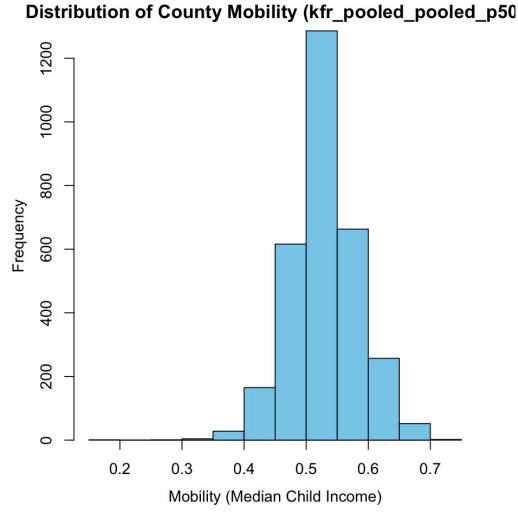
Mass emigration of highly skilled individuals looking for better opportunities in other countries



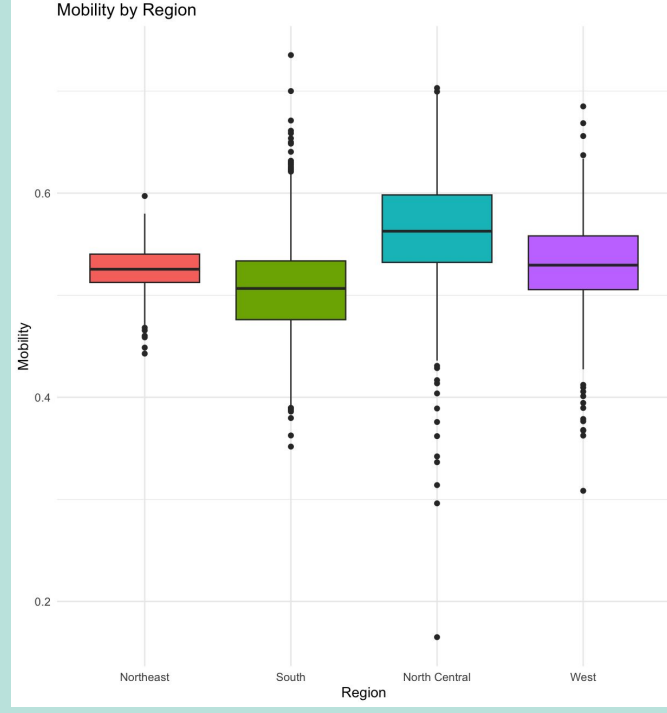
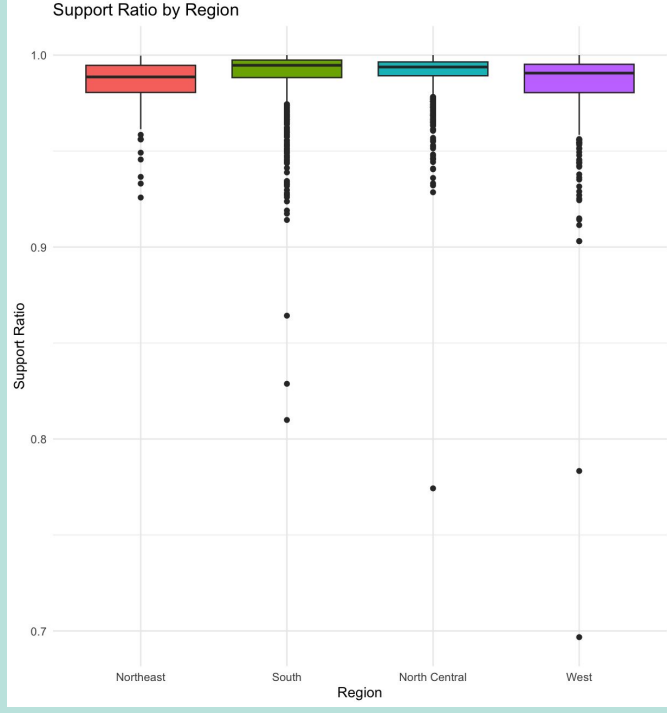
Also known as human capital flight. There is much nuance pertaining to this concept and its consequences.

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Overview of Data (cont.) Why kfr_pooled_pooled_p50? — Considered as a widely accepted measure of intergenerational economic mobility, representing the average income rank in adulthood (around age 35) of children whose parents were at the 25th percentile of the national income distribution, pooled across genders and cohorts. Offers a clear, comparable indicator of upward mobility, allowing researchers to assess how well children from low-income families can move up the economic ladder in different counties.		
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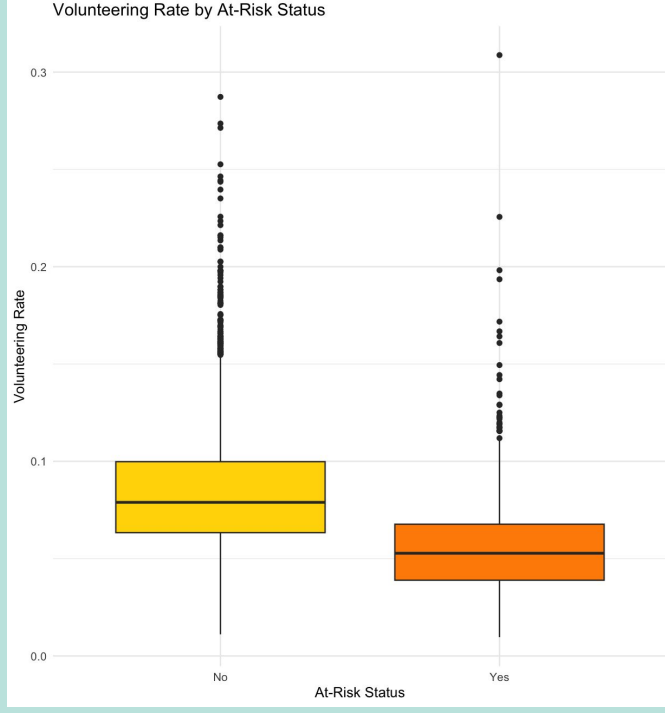
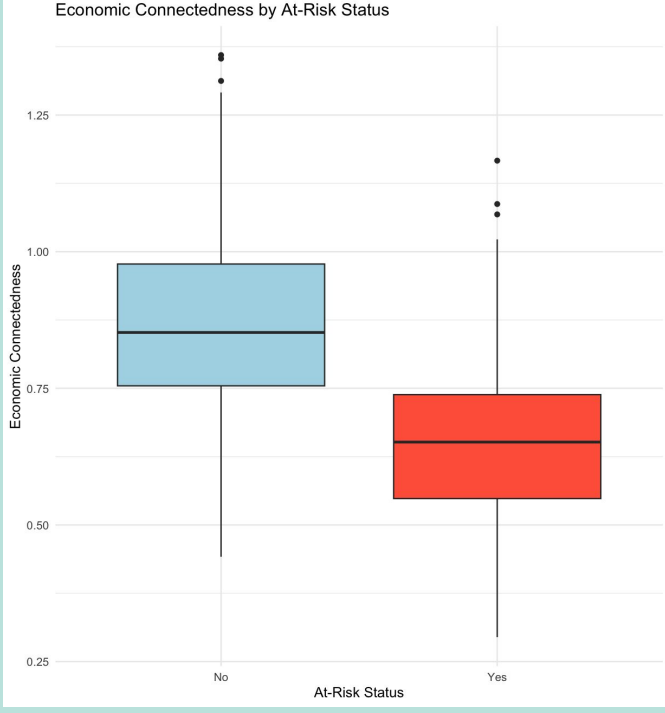
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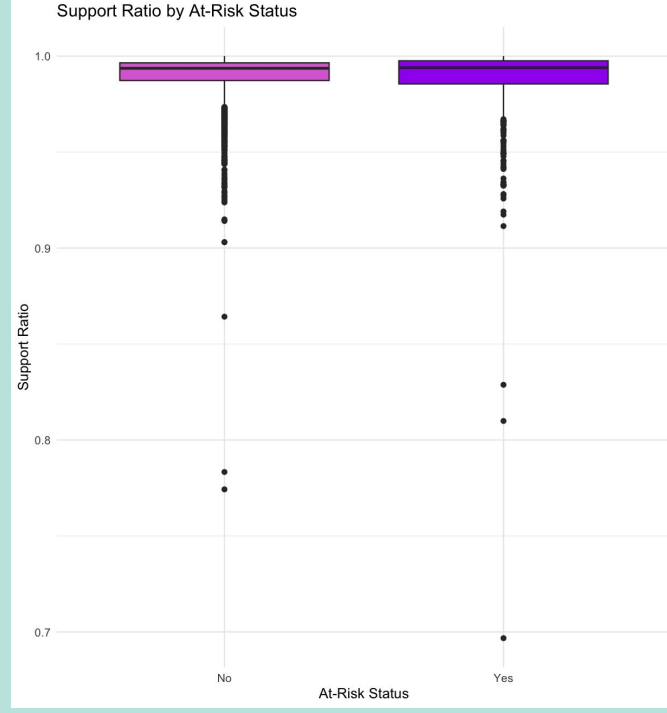
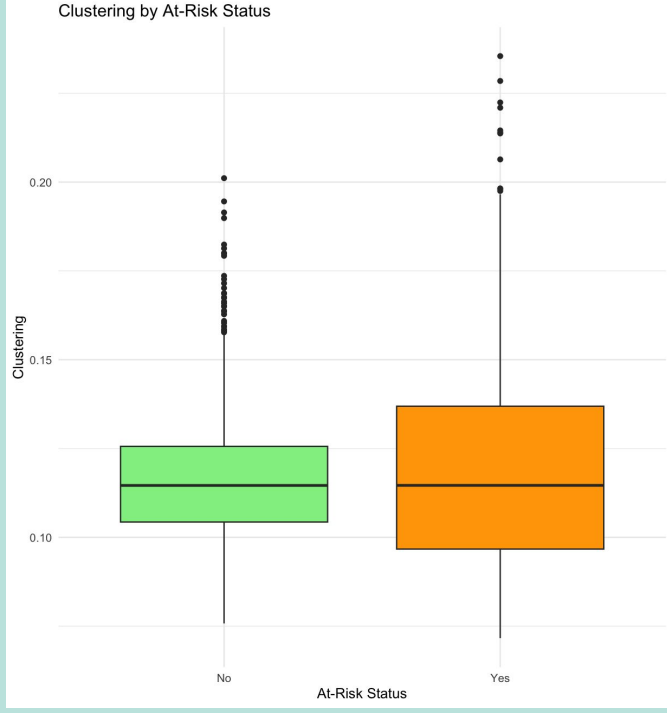
EDA



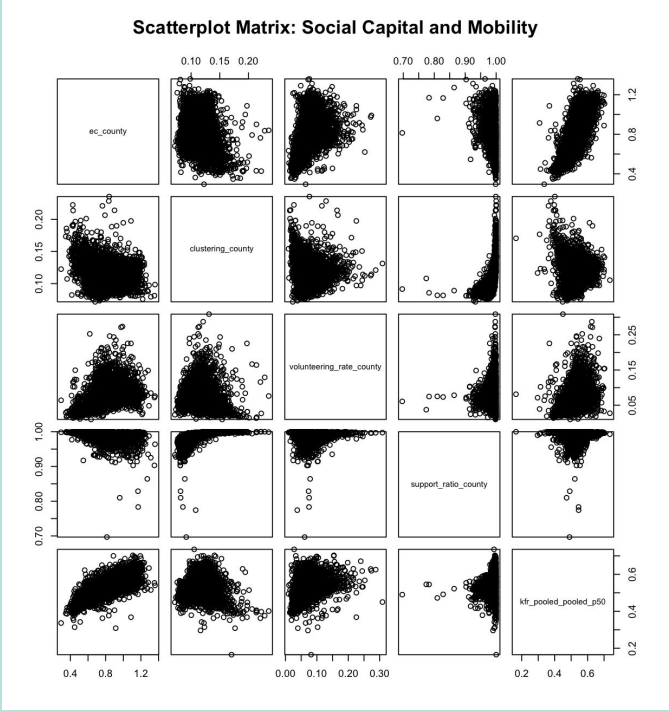
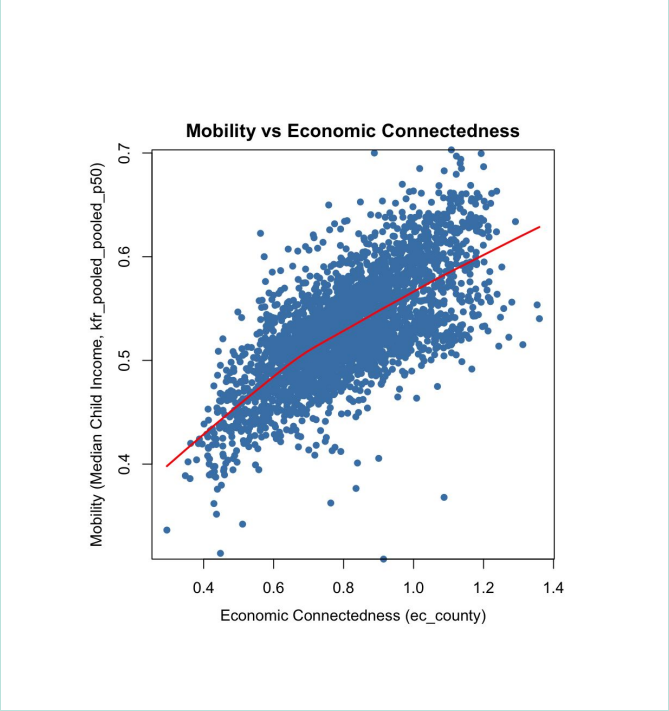
EDA



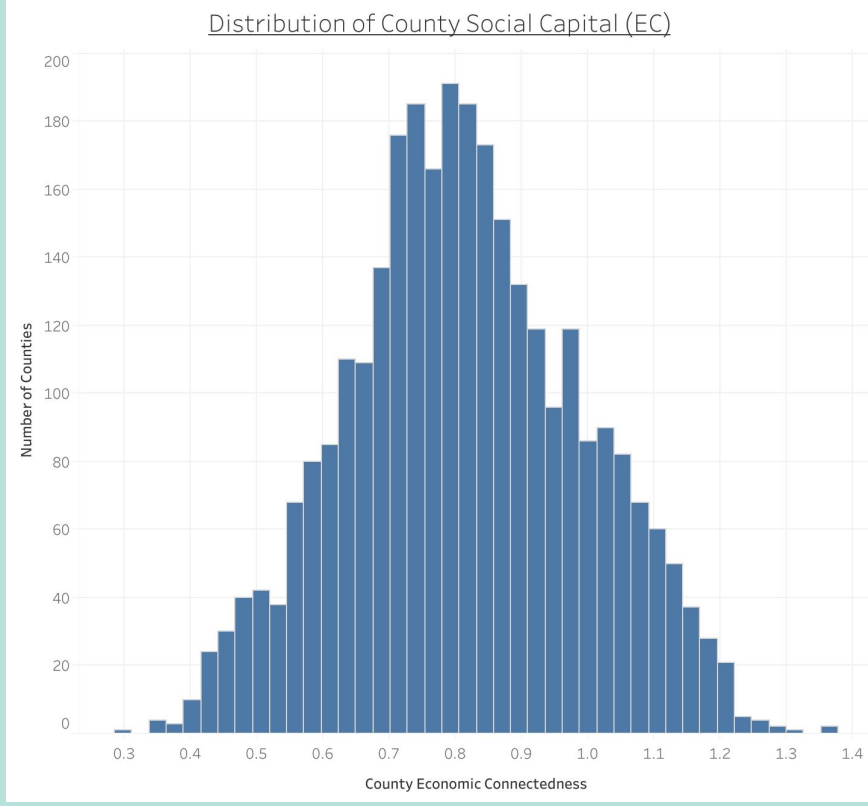
EDA



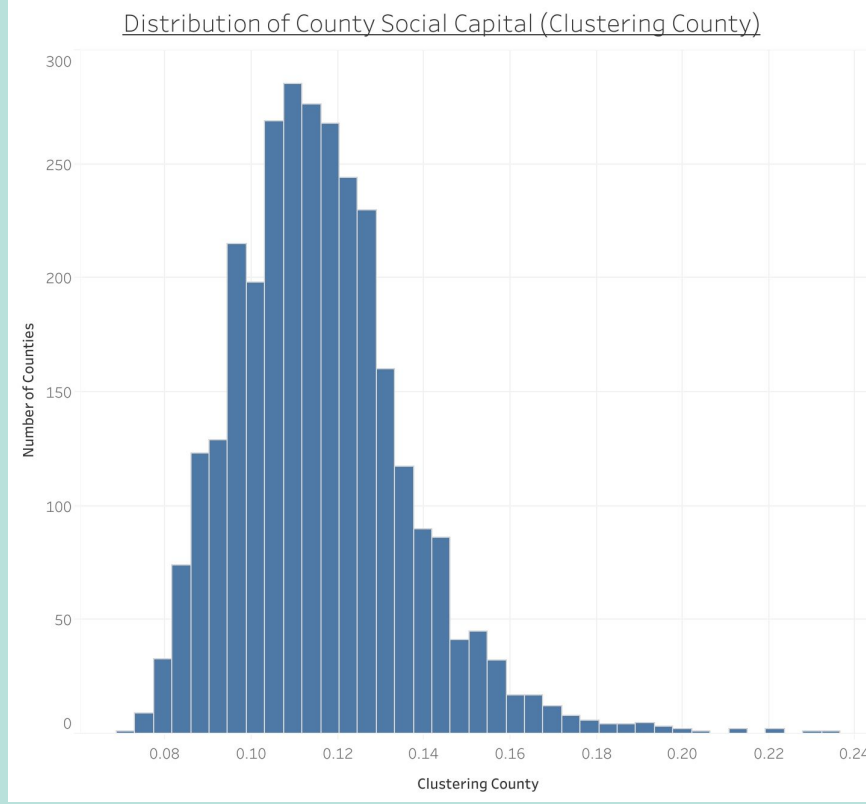
EDA



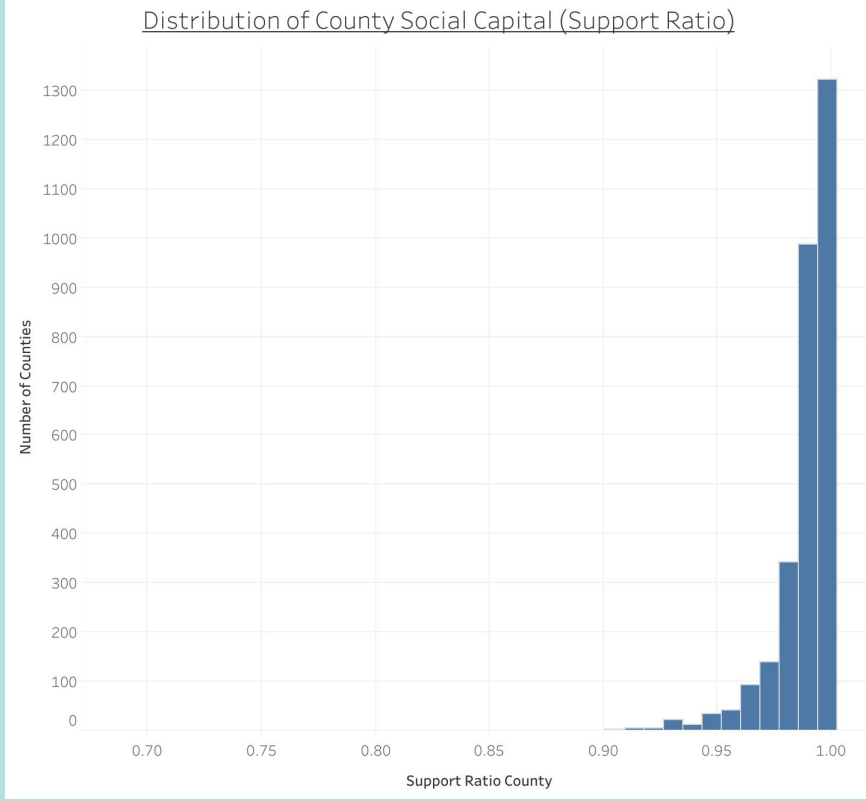
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EDA

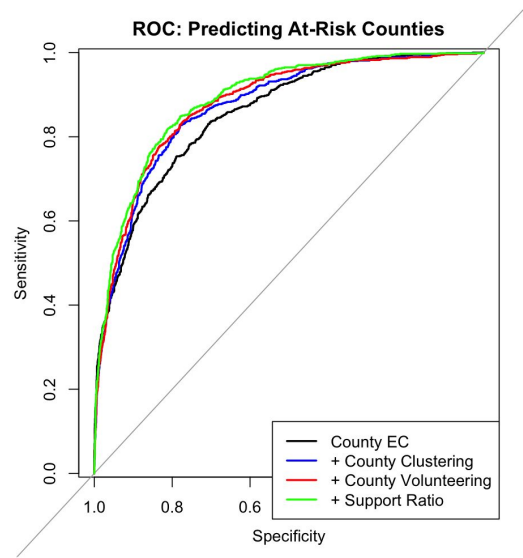


EDA



Models

Logistic Regression



AUCs:

County EC (only) - 0.854

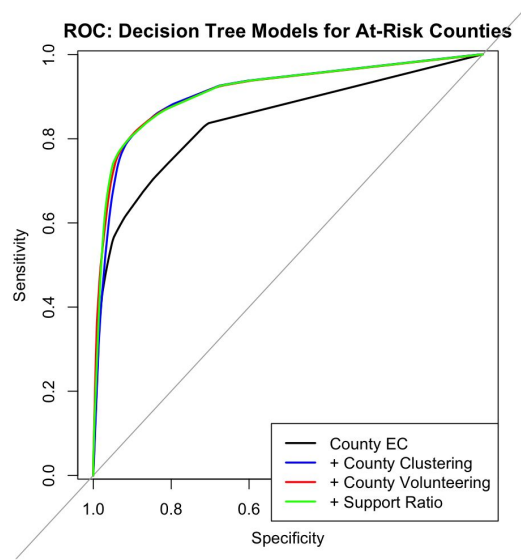
+ County Clustering - 0.871

+ County Volunteering - 0.879

+ Support Ratio - 0.889

Models

Decision Trees



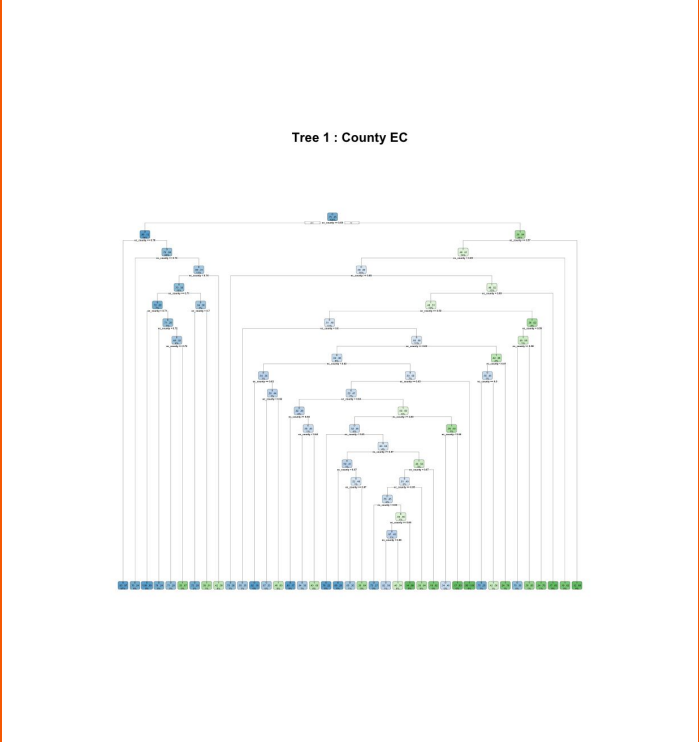
AUCs:

- County EC (only) - 0.842
- + County Clustering - 0.909
- + County Volunteering - 0.913*
- + Support Ratio - 0.913*

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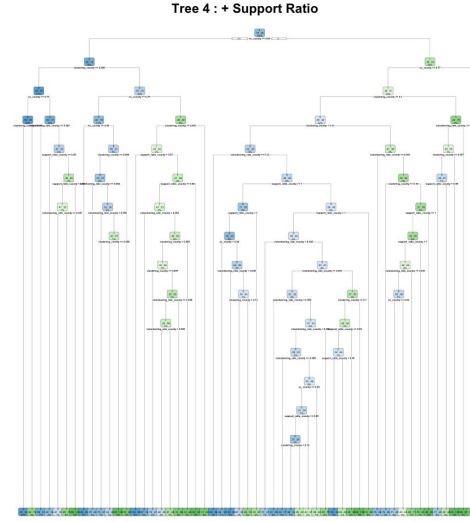
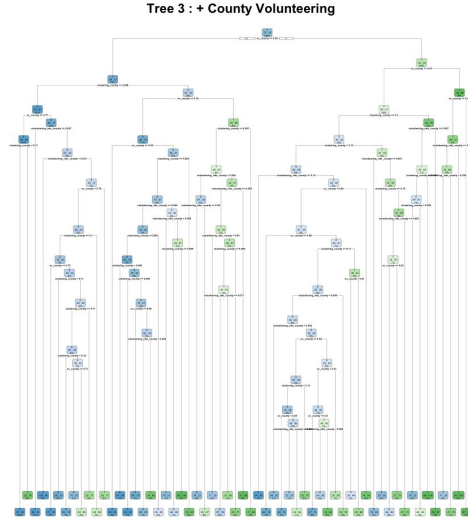
Models

Decision Trees

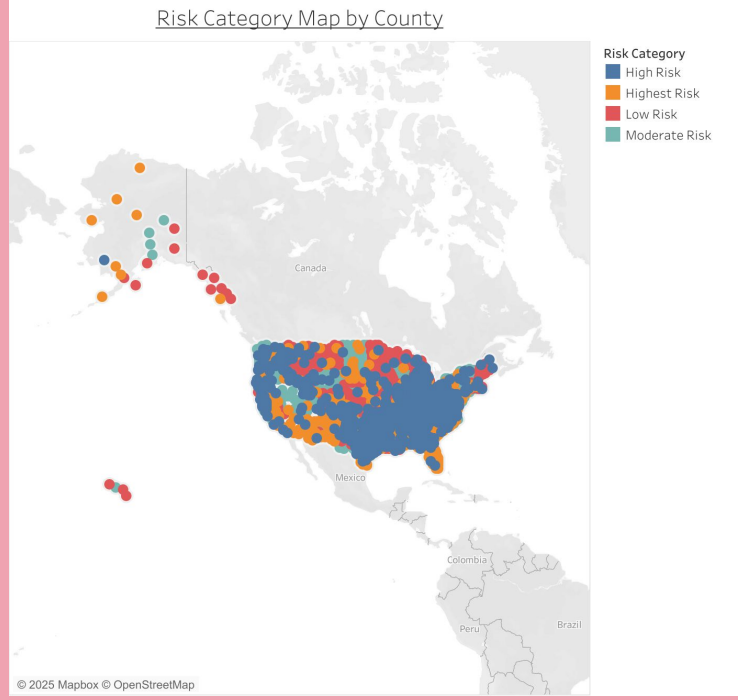


Models

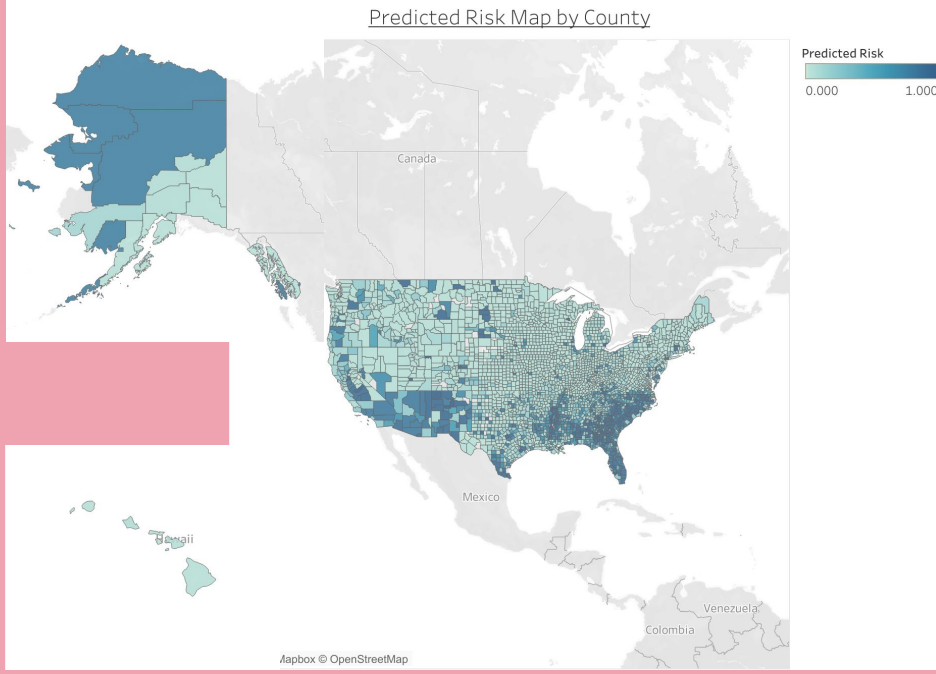
Decision Trees



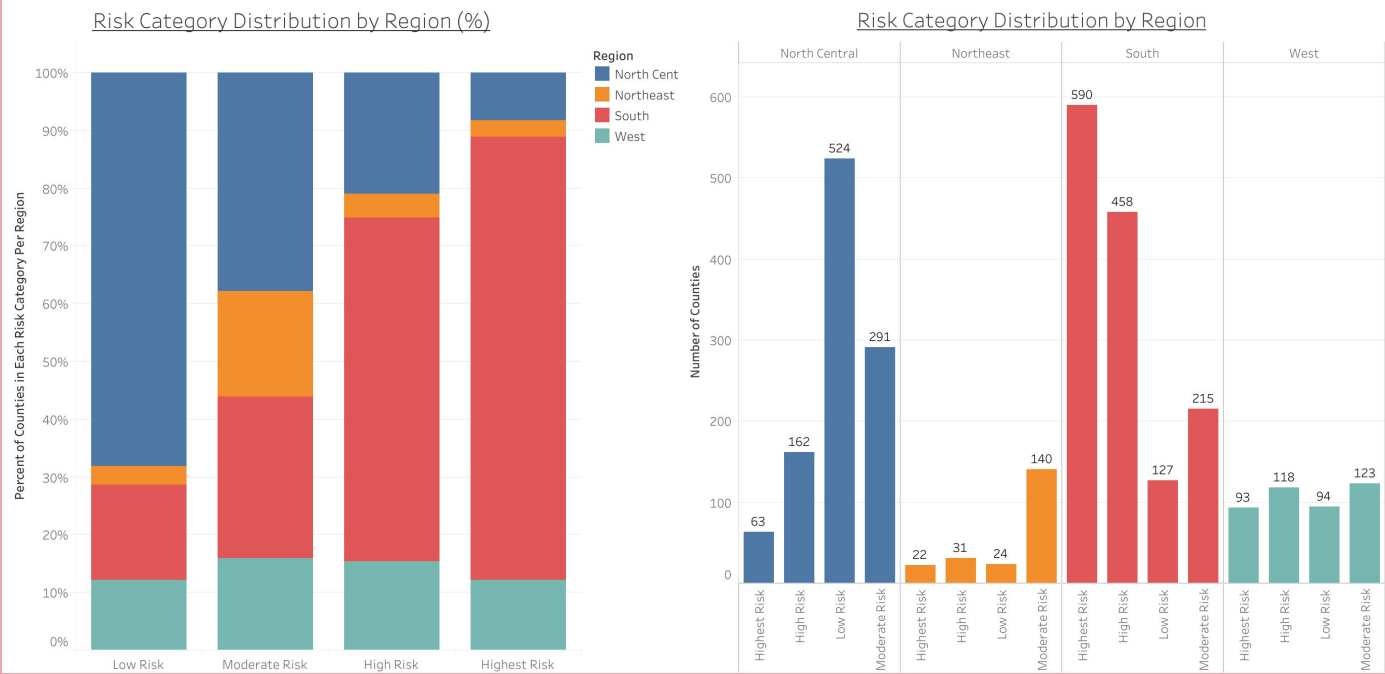
Discussion



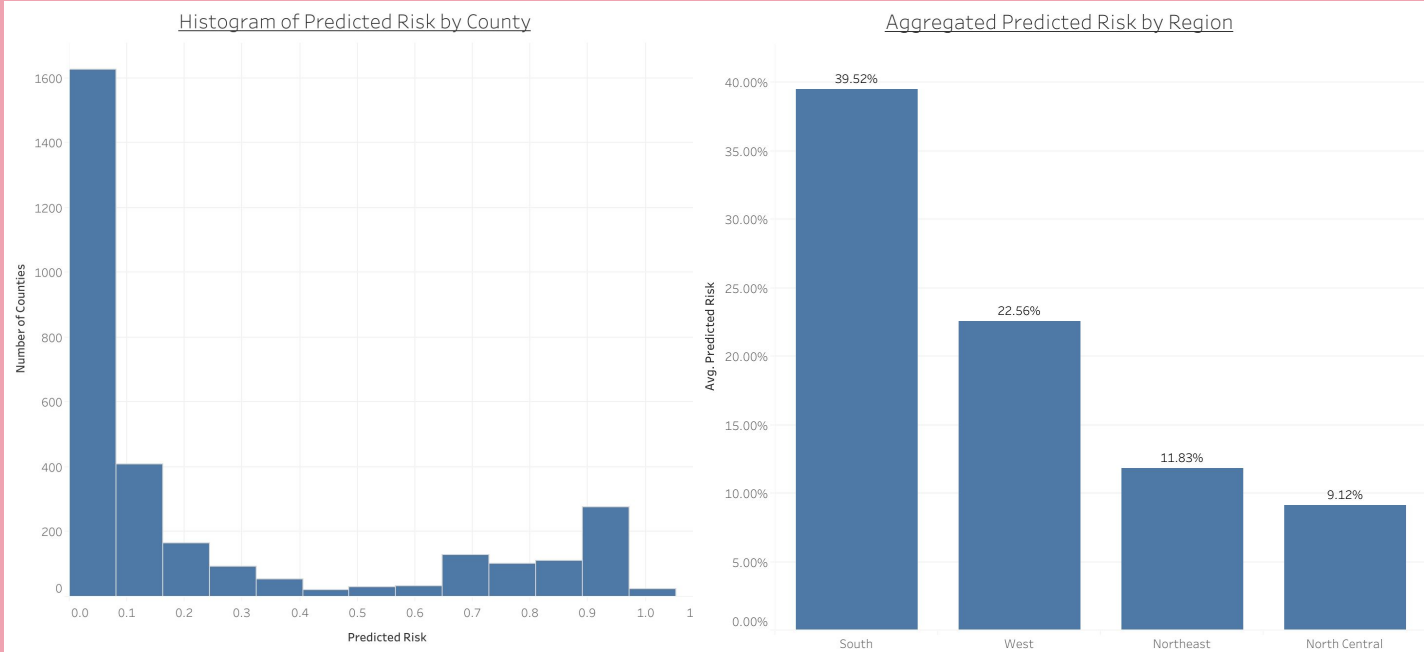
Discussion



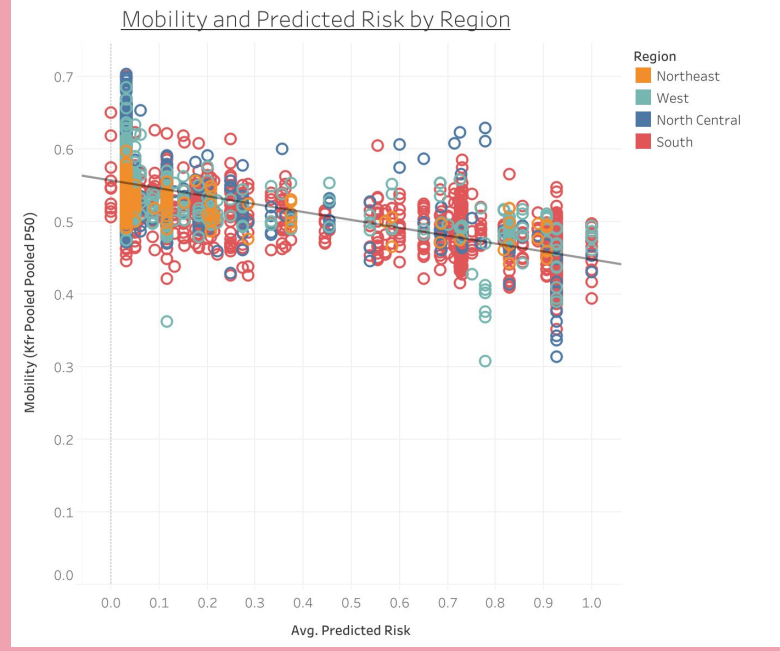
Discussion



Discussion



Discussion



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Limitations

These findings should be interpreted with caution, as the model is constrained by the availability and quality of county-level data and may not capture all local or emerging factors influencing migration. Important cultural, demographic, or policy influences may be underrepresented, and within-region variation means that local context remains essential for effective intervention.

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Thank you.

Mercy Eme

Professor Barr

STATS 418

“Brain Drain”: Identifying U.S. Regions at Risk for Future Capital Flight Using Social Capital and Mobility Outcomes

Background

Japa: “to flee” or “to escape” in Yoruba. More commonly known as “brain drain” or human capital flight, this phenomenon is becoming increasingly prevalent not only among young Nigerians, but also among people worldwide who are eager to seek new opportunities for advancement. Driven by a desire to escape economic uncertainty and limited prospects at home, this concept describes the mass exodus of intelligent, highly ambitious, and highly skilled individuals from their hometown, state, or country.

Even within first-world countries like the United States, internal “brain drain” is transforming regions, as talented professionals migrate from rural towns to vibrant urban centers like New York City or San Francisco in search of better prospects. Such migration can profoundly affect the communities left behind, leading to declines in civic engagement, weakened community ties, and increased strain on essential services.

Overview of Data

The Opportunity Atlas, developed by the U.S. Census Bureau and Opportunity Insights at Harvard, provides detailed data on neighborhood-level mobility and long-term outcomes for children. This project utilizes data from the Atlas to analyze and predict the risk of future human capital flight across U.S. counties. The Atlas identifies areas where children, particularly those from disadvantaged backgrounds, face limited upward mobility and are more likely to leave in search of better prospects.

Multiple datasets inform this analysis, including Social Capital Data by County, County By Cohort Estimates, and County Names and FIPS. These datasets provide insight into population characteristics, social networks, civic engagement, and movement patterns of individuals from various income backgrounds. The County By Cohort Estimates dataset serves as the primary source for assessing how childhood environment and parental income shape economic outcomes at the county level. Standardization of county identifiers (FIPS codes) was necessary for accurate merging across datasets.

The main mobility metric from the County By Cohort Estimates dataset, *kfr_pooled_pooled_p50*, represents the mean percentile rank of household income at age 27 for

children from median-income families, pooled across race and gender. This measure offers a clear, comparable indicator of upward mobility, avoiding extremes and highlighting regions with systemic barriers to economic advancement. U.S. counties with low values on this metric are identified as at higher risk for future brain drain, as young adults are more likely to leave these areas in pursuit of greater opportunities.

EDA

The exploratory data analysis assessed relationships among social capital indicators (economic connectedness, clustering, volunteering rate, and support ratio) and county-level economic mobility. Economic connectedness showed a strong positive correlation with both volunteering rates and intergenerational mobility ($r = 0.69$), while volunteering rates were also moderately correlated with mobility ($r = 0.33$). Clustering was negatively related to economic connectedness and moderately associated with support ratio.

Mobility scores across counties followed a roughly normal distribution centered near 0.5. Using a model-driven threshold, about 25% ($No = 0.2500813$) of counties were classified as “at-risk” for low mobility. At-risk counties generally exhibited lower economic connectedness and volunteering, with slightly higher clustering, while support ratios remained high overall but were modestly lower in these areas. These patterns indicate that deficits in social capital, especially in economic connectedness and civic engagement, are linked to lower mobility prospects.

Regional variation was observed, with Southern counties tending to have lower mobility and support ratios. Overall, these findings reinforce the role of social capital in upward mobility and highlight areas for further analysis and policy focus.

Modeling and Additional Analysis

To predict counties at risk for low economic mobility, a series of nested logistic regression and decision tree models were constructed using county-level social capital indicators. Each model incrementally added predictors: starting with economic connectedness, then including network clustering, volunteering rate, and support ratio. All models used a binary outcome indicating at-risk status.

Results from the logistic regression models showed that higher economic connectedness, lower clustering, greater volunteering rates, and higher support ratios were each associated with reduced risk. Model performance, as measured by the area under the ROC curve (AUC), improved with the inclusion of each additional variable. The AUC increased from 0.85 (economic connectedness only) to 0.89 (all variables included).

Similarly, decision tree models were fit using the same sequence of predictors. These models demonstrated high discriminatory power, with AUC values rising from 0.84 to 0.91 as additional variables were included. The model, tree model 3, incorporating economic

connectedness, clustering, and volunteering rate (AUC = 0.91) balanced predictive accuracy and interpretability, and was selected for generating risk predictions across all counties.

Predicted at-risk probabilities from the selected decision tree model were assigned to each county and divided into quartiles for visualization and further analysis.

Discussion and Recommendations

The predictive modeling results reveal regional disparities in brain drain risk across U.S. counties. Notably, the South emerges as the region with the highest average predicted risk, nearly doubling the figures observed in the West and far surpassing those in the Northeast and North Central regions. This pronounced gap indicates that counties in the South are, on average, significantly more susceptible to the drivers of brain drain identified by the model.

Examining the relationship between predicted risk and actual economic mobility, the data show a clear negative association: counties with higher predicted risk scores tend to exhibit lower levels of upward mobility. This trend is particularly evident in the South, where many counties cluster in the high-risk, low-mobility quadrant. The alignment between model predictions and observed mobility outcomes supports the utility and validity of the risk model.

Further analysis of the distribution of risk categories across regions underscores the concentration of vulnerability. The South dominates the “High Risk” and “Highest Risk” categories, while the North Central and Northeast regions are more prevalent in the “Low Risk” and “Moderate Risk” groups. The West features a more even distribution but still contains notable clusters of high-risk counties. This suggests that interventions aimed at reducing brain drain should be regionally tailored, with a particular focus on the South.

Overall, the histogram of predicted risk highlights that while the majority of U.S. counties face relatively low risk, there is a distinct subset experiencing much higher levels of vulnerability. The spatial maps further emphasize these patterns, showing clusters of high-risk counties especially in the Southeast, the West, and Alaska. Collectively, these visualizations demonstrate that the threat of brain drain is not uniformly distributed but is instead highly concentrated in particular regions (most notably the South) where targeted policy action is likely to have the greatest impact. Targeted investments in education, workforce development, and local economic opportunity may help retain talent and boost mobility in these vulnerable areas. Additionally, strengthening social capital, such as supporting local civic organizations and networks, could buffer against the factors driving out-migration.

Limitations

These findings should be interpreted with caution, as the model is constrained by the availability and quality of county-level data and may not capture all local or emerging factors influencing migration. Important cultural, demographic, or policy influences may be underrepresented, and within-region variation means that local context remains essential for effective intervention.

References

Chetty, Raj, et al. (2022a). “Social Capital I: Measurement and Associations with Economic Mobility.” *Nature*, 608(7921), 108–121.

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